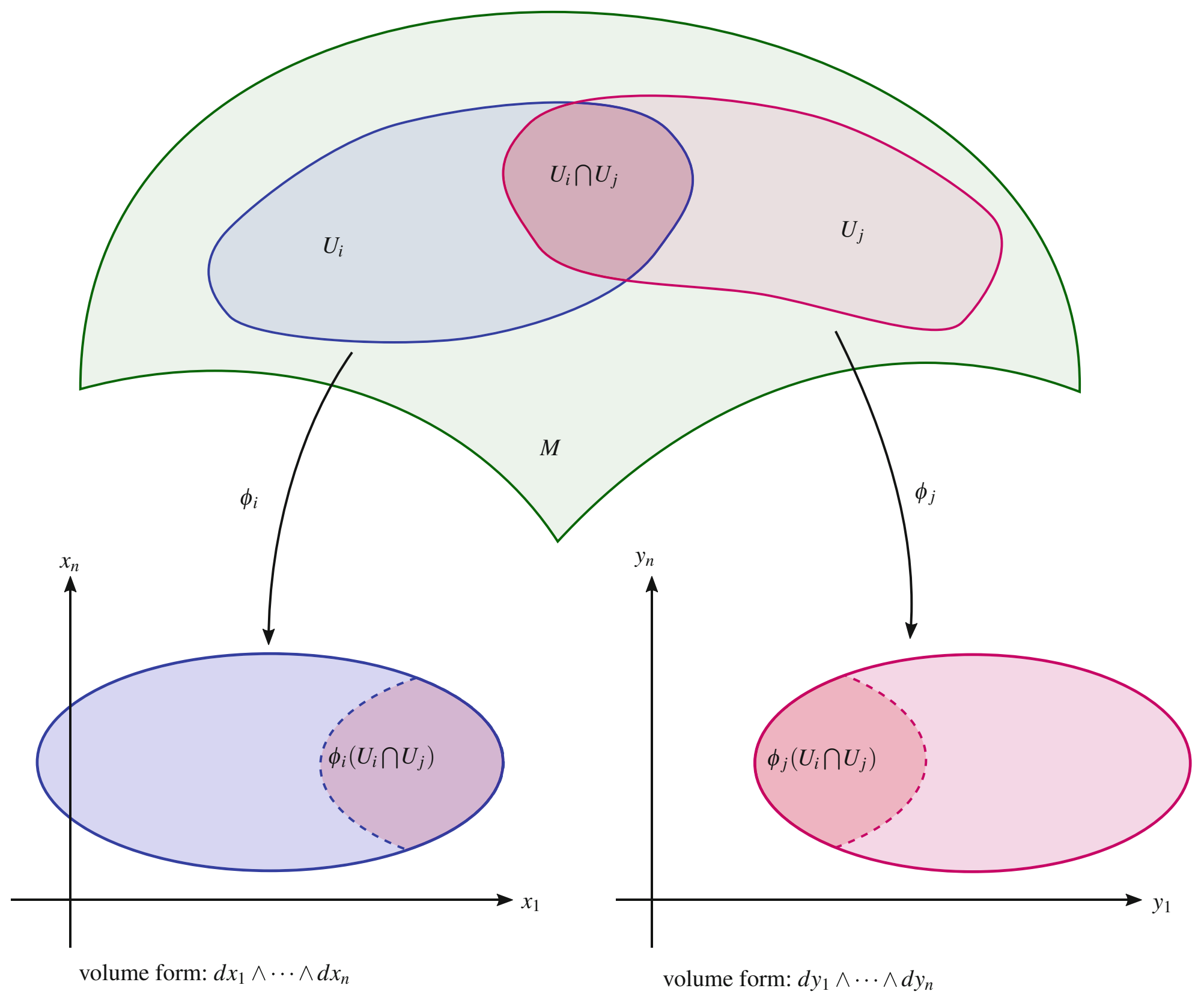

$U_i \cap U_j$
U_i
U_j
M
ϕ_i
ϕ_j
x_n
x_1
$\phi_i(U_i \cap U_j)$
y_n
y_1
$\phi_j(U_i \cap U_j)$
volume form: $dx_1 \wedge \cdots \wedge dx_n$
volume form: $dy_1 \wedge \cdots \wedge dy_n$